FLAIR: A Federated ICU Benchmark Using the Common Longitudinal ICU Format

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Abstract

Background. AI clinical decision support tools are proliferating faster than the evidence needed to judge whether they work across hospitals. ICU models are often evaluated at one or two sites, or on a small number of public electronic health record datasets, rather than tested broadly across independent hospitals. Public benchmarks are essential for method development, but repeated tuning on the same datasets can produce benchmark saturation and illusory generalizability. Traditional multi-site validation requires bilateral data use agreements and patient-level data transfer, so it often takes years and reaches only a few sites.

Methods. FLAIR is an open-source federated benchmark and model-to-data protocol for ICU prediction research. It uses the Common Longitudinal ICU Format (CLIF), a critical-care data format with shared table structures and mCIDE categories. Developers build and test code on public MIMIC-CLIF data. After Benchmark Committee review, the same code or container can run locally at participating CLIF sites. Five binary benchmark tasks are included: daily in-hospital mortality, daily long-term acute care hospital discharge, extubation failure within 24 hours, sepsis within 6 hours, and unplanned ICU readmission. Each site returns aggregate report bundles for cohort char-

acteristics, discrimination, calibration, decision curves, lead time, subgroup performance, and audit metadata. Patient-level rows remain at the site.

Results. In the v2 baseline runs, an XGBoost count-feature model trained on MIMIC-CLIF had internal-test AUROCs of 0.864 for daily mortality, 0.789 for LTACH discharge, 0.696 for extubation failure, 0.693 for sepsis, and 0.641 for ICU readmission. At the first external reporting site, Rush, AUROC was lower for all five tasks: 0.748, 0.658, 0.565, 0.407, and 0.509, respectively. Site-level results for additional institutions remain TODO. **TODO:** Confirm or regenerate T4 sepsis results against the current CDC Adult Sepsis Event task implementation before submission.

Conclusions. FLAIR is intended to make external validation more reproducible and more practical for ICU AI. It provides public development data, inspectable task definitions, committee-reviewed model-to-data execution, and aggregate-only reporting for a growing leaderboard. The current release is a benchmark and protocol resource, not evidence of clinical utility for any specific model.

1 Introduction

The bottleneck in ICU clinical AI is no longer the supply of models. Predictive algorithms for sepsis, acute kidney injury, extubation readiness, ICU mortality, and hospital operations continue to appear in the literature. The harder problem is evidence: whether a given model behaves acceptably in hospitals that did not build it. That evidence gap is partly an access gap. Representative, multi-institutional ICU data are usually not available to the team that needs to test a model.

External validation of flagship electronic health record-based clinical decision support has remained narrow. The Epic Sepsis Model, used across many United States hospitals, was first independently evaluated at one site, where discrimination fell to an AUC of 0.63 against vendor-claimed ranges of 0.76 to 0.83 (**??**). Duke's Sepsis Watch required years before external assessment at another site (**?**). The Targeted Real-time Early Warning System

reached five sites within one health system (**?**). Mayo Clinic's electrocardiogram-based low-ejection-fraction screen, developed through a regulatory pathway, required multi-year coordination to cover several external sites (**?**). These examples show that serious validation is possible, but the process does not scale easily to the pace of model development.

Public benchmarks have helped but cannot solve the problem alone. ICU benchmarks built on MIMIC, eICU, HiRID, AUMCdb, and related datasets are essential for reproducible development (**??**). They also inherit the selection biases of their source data (**?**). Repeated tuning against a fixed public benchmark can make performance appear more transportable than it is (**??**). Private multi-site validation addresses that weakness, but the usual route - data use agreements followed by transfer of de-identified patient-level data to an analytic center - is slow and narrow.

Several communities have shown pieces of a better approach. OHDSI and LEGEND federate studies across the OMOP common data model (**?**). MedPerf demonstrates portal-orchestrated model-to-data benchmarking for medical imaging (**?**). The FeTS Challenge used federation for imaging-based training rather than validation (**?**). OpenSAFELY showed that code can travel to national-scale clinical data without moving patient rows (**?**). DREAM Challenges introduced model-to-data challenge infrastructure, and the EHR DREAM Challenge showed that this pattern can support EHR-based mortality prediction (**??**). What has been missing for ICU EHR-based clinical decision support is the combination of a critical-care data standard, a reusable task catalog, and a standing federated validation protocol.

FLAIR (Federated Learning Assessment for ICU Research) was built to fill that gap. The core idea is simple: the model goes to the data, and aggregate evaluation reports come back. A developer writes code against MIMIC-CLIF, the public CLIF version of MIMIC-IV (**??**). After review, the same code can run inside each participating institution. The site returns tables and metrics, not patient rows.

This paper describes FLAIR as a dataset, benchmark, and protocol resource. The dataset layer is CLIF, with MIMIC-CLIF as the public development entry point. The benchmark layer

is a seed set of five ICU prediction tasks implemented as inspectable Python cohort builders. The protocol layer is a repeatable model-to-data workflow with a Benchmark Committee, a FLAIR Portal, local execution by Site Leads and Data Stewards, and aggregate-only reporting to a leaderboard. FLAIR is not a clinical deployment system and does not certify models for bedside use. Its purpose is narrower: to make multi-site ICU model evaluation reproducible, auditable, and feasible without routine transfer of patient-level data.

2 Methods

FLAIR is implemented as the `flair-benchmark` Python package and command-line interface. The package builds task cohorts from CLIF tables, provides optional point-in-time feature extraction, and creates aggregate report bundles from prediction files. Each cohort row has a `prediction_dttm`, which is the no-future-data boundary. Models may use data at or before that timestamp and no data after it.

We describe FLAIR as three coupled artifacts: a data layer, a task layer, and a protocol layer. The data layer is CLIF, with MIMIC-CLIF as the public development entry point. The task layer is a set of benchmark definitions implemented as single-file Python modules. The protocol layer is a model-to-data execution process in which reviewed code or containers run locally at each site and return aggregate JSON reports.

The public workflow has five steps: configure a CLIF site, build a task cohort with `flair load-task`, optionally extract features with `flair fe-wide` or `flair fe-meds`, train or apply a model, and create aggregate reports with `flair build-report`. Binary prediction files carry `prediction_id`, the task label, split, and `y_prob`. The report builder scores the held-out test split and writes JSON files that can be pooled centrally.

Reporting follows the general structure expected for clinical prediction models, including discrimination, calibration, threshold behavior, and subgroup analysis (**?**). The FLAIR report bundle populates many of the elements expected by TRIPOD+AI. A Datasheet for Datasets

framework (**?**) informs the public MIMIC-CLIF cohort documentation, and a Model Card framework (**?**) informs the baseline model documentation.

Institutional review. Each participating consortium site obtained independent institutional review board approval for CLIF participation and local execution of FLAIR benchmark code. Central protocol coordination is provided by the University of Chicago. **TODO:** Confirm final IRB wording for all sites.

Data availability. MIMIC-CLIF is publicly accessible through PhysioNet under the applicable MIMIC data use agreement. Consortium site data are not redistributable. They are accessible only through approved local execution at participating sites.

Code availability. The FLAIR source code, task definitions, feature extractors, report generators, and CLIF-WorkBench/container specifications are released under Apache 2.0 at `https://github.com/clif-consortium/FLAIR`. Baseline model archive DOI: **TODO:** DOI. Container registry tag: **TODO:** tag.

3 Datasets

The Common Longitudinal ICU Format

The Common Longitudinal ICU Format is a critical-care data format that harmonizes ICU electronic health record content into a limited set of relational tables with mCIDE category definitions (**?**). FLAIR uses CLIF tables for patients, hospitalizations, admission-discharge-transfer events, vitals, labs, medications, respiratory support, patient assessments, microbiology, diagnoses, and other domains where needed. The current code loads CLIF through `clifpy` and converts data to local wall-clock timestamps before cohort construction. CLIF is useful for benchmarking because the task code can refer to common table names

and mCIDE category values. This does not remove all site differences, but it reduces the site-specific feature engineering needed before a model can be evaluated. In spirit, CLIF is to critical care what mCODE is to oncology: a clinically scoped common-data-element layer that makes downstream analytic code more portable (**?**). CLIF is complementary to broader EHR machine-learning standards such as the Medical Event Data Standard (**?**). FLAIR can consume CLIF relational tables directly, and future bridges can express CLIF data as event streams for compatible model pipelines.

MIMIC-CLIF

MIMIC-CLIF is the public development dataset. It expresses MIMIC-IV critical care data in the CLIF v2.1 schema and contains approximately 70,000 ICU stays (**??**). Developers can use MIMIC-CLIF to debug task definitions, train baselines, and test whether their code follows the cohort and prediction-file contracts before federated execution. This creates a develop-on-public, validate-on-private workflow without requiring private patient-level data to leave consortium sites.

Federated Site Data

Consortium data are not redistributed. Each site maps its own ICU data to CLIF, runs approved FLAIR code locally, and returns aggregate JSON reports. Across CLIF sites, FLAIR is intended to cover heterogeneous academic medical centers and health systems. Exact consortium-level site counts and cohort totals should be finalized at submission. **TODO:** Confirm final site count, hospital count, ICU bed count, and aggregate ICU stay count.

Table **??** shows the current federated cohort skeleton from the v2 manuscript. MIMIC-CLIF is populated as the public development entry point. Rush is the first reporting external site. Additional site columns remain TODO until federated runs are complete.

Characteristic	MIMIC-CLIF	Rush	Site C	Site D
ICU stays	72,778	13,839	TBD	TBD
Female (%)	43.2%	47.5%	TBD	TBD
Age, median (yr)	67	65	TBD	TBD
SOFA hospital maximum, median	5	4	TBD	TBD

Table 1: Federated consortium cohort skeleton for the T1 cohort. MIMIC-CLIF is the public development entry point. Rush is the first reporting external site. Remaining site columns are TODO. Cell suppression at $n < 10$ is applied site by site. Source: FLAIR v2 baseline artifacts and site aggregate reports.

Measure	T1 mortality	T2 LTACH	T3 extubation	T4 sepsis	T5 readmission
ICU encounters	64,408	64,408	26,768	80,600	67,235
Decision rows	198,125	198,125	26,771	14,347,321	67,250
Positive rate, row level	19.2%	14.3%	5.4%	0.2%	9.2%
Stays ever positive	13.0%	6.3%	5.4%	7.2%	9.2%
Sex, F/M	27,795/36,613	27,795/36,613	10,031/16,737	35,760/44,840	29,199/38,036
Age 65 years or older	55.3%	55.3%	54.1%	54.8%	54.1%
SOFA hospital maximum, median (IQR)	5 (2–8)	5 (2–8)	7 (5–10)	4 (2–7)	4 (2–7)
Charlson score, median (IQR)	2 (0–4)	2 (0–4)	2 (0–4)	2 (0–4)	2 (0–4)

Table 2: MIMIC-CLIF cohort snapshot per task from the v2 baseline artifacts. Decision rows are one per stay for T3 and T5, daily at 07:00 local for T1 and T2, and hourly for T4. Counts are per stitched encounter, using `hospitalization_join_id`. Charlson and Elixhauser scores are missing for 49–54% of stays in the MIMIC-CLIF export. Source: FLAIR v2 baseline artifacts. **TODO:** Regenerate this table after final task-code freeze, especially for T4.

Privacy-Preserving Access

Three properties define the FLAIR access model. First, no patient-level data leave any site; outputs are aggregate JSON reports produced by the FLAIR library inside local execution. Second, the library enforces suppression rules: counts below 10 in cohort tables are suppressed; demographic subgroups with fewer than 10 observations are omitted from subgroup reports; and calibration bins with fewer than 10 observations do not report observed event rates. Third, governance review occurs before execution. The Benchmark Committee reviews submitted code and models for compatibility, clinical face validity, and security concerns before any artifact is distributed to sites. Sites retain final approval over which reviewed submissions run locally.

4 Benchmarks

Task Taxonomy

Every FLAIR task is described by metadata and implemented as code. The `META` dictionary records the task class, temporal pattern, prediction unit, label column, clinical threshold, threshold direction, and recommended report mode. The `build()` function returns a cohort parquet with one row per prediction opportunity. Every row includes a stable `prediction_id`, a stitched encounter key, a prediction timestamp, the task label, train/test split, and demographic variables used for reporting. The design keeps the boundary between a new task and an existing task explicit. Reviewers can inspect the metadata before reading the implementation.

Initial Five-Task Suite

The v1 release covers five binary ICU tasks spanning mortality, disposition, respiratory support, infectious deterioration, and ICU utilization.

- **T1. Daily in-hospital mortality.** At local 07:00 during the first ICU stay, predict whether the hospitalization will end in death or hospice discharge. The current implementation uses an unconditional adult ICU cohort; there is no discharge-disposition filter at cohort entry. The recommended report mode is landmark.
- **T2. Daily LTACH discharge.** At the same daily 07:00 anchors as T1, predict eventual discharge to a long-term acute care hospital. T1 and T2 share the same cohort and grid, but they are not complementary labels. A home discharge is negative for both. The recommended report mode is landmark.
- **T3. Extubation failure within 24 hours.** At the first confirmed extubation after invasive mechanical ventilation, predict death or reintubation within 24 hours. The

implementation uses a consensus-window ventilation episode detector to avoid treating short device-status artifacts as true intubation or extubation events. The task targets the respiratory liberation question (**?**). The recommended report mode is episodic.

- **T4. Sepsis within 6 hours.** On an hourly hospital-wide grid, predict CDC Adult Sepsis Event onset within the next 6 hours. The current implementation uses the CLIF Adult Sepsis Event utility, includes lactate, requires the CLIF `microbiology_culture` table, and stops predictions strictly before the sepsis onset timestamp. This replaces older draft wording that no longer matches the current task code. The recommended report mode is peak. **TODO:** Confirm all baseline results and captions after this T4 definition is frozen.

- **T5. Unplanned ICU readmission.** At first ICU discharge, predict return to the ICU later in the same hospitalization through an unplanned path. The label is derived from the ADT location stream. Planned ICU-to-procedure-to-ICU movements are merged and not counted as readmissions. This framing follows structured-EHR ICU-readmission prediction (**?**). The recommended report mode is episodic.

Count-Based Reference Baseline

The reference baseline is an XGBoost model over count features. CLIF events are converted to deterministic tokens in the EHRSHOT/FEMR tradition (**??**). Events include vitals, labs, medications, respiratory support, patient assessments, and other configured domains. Each prediction row is represented by counts of events observed before `prediction_dttm`. Missing tokens are zero. Tokens not seen during training are ignored at inference. No task-specific feature curation is used in the reference baseline.

The model is trained on MIMIC-CLIF and then frozen for external evaluation. The same trained model and token dictionary are shipped to sites. Per-task clinical thresholds, alarm directions, and report modes are single-sourced in the package policy registry and folded into task audit metadata, so a policy change is visible at every site.

Table **??** reproduces the v2 headline baseline results. These values should be read as benchmark checks, not as a claim that the baseline is clinically adequate. The main finding is cross-site degradation: all five tasks performed worse at Rush than on the MIMIC-CLIF internal holdout.

Task	Mode	MIMIC-CLIF AUROC (95% CI)	Rush AUROC (95% CI)	Site C	Site D
T1, ICU daily mortality	landmark	0.864 (0.856–0.872)	0.748 (0.735–0.763)	TBD	TBD
T2, LTACH discharge	landmark	0.789 (0.775–0.803)	0.658 (0.622–0.691)	TBD	TBD
T3, extubation failure 24 h	episodic	0.696 (0.665–0.719)	0.565 (0.522–0.605)	TBD	TBD
T4, sepsis 6 h	peak	0.693 (0.678–0.709)	0.407 (0.391–0.421)	TBD	TBD
T5, unplanned ICU readmission	episodic	0.641 (0.626–0.658)	0.509 (0.489–0.531)	TBD	TBD

Table 3: Federated leaderboard skeleton for the XGBoost count-feature baseline. AUROC values are mode matched and include 95% bootstrap confidence intervals from the v2 manuscript. MIMIC-CLIF is the internal holdout. Rush is the first external reporting site. Remaining site columns are TODO. Source: FLAIR v2 baseline artifacts and Rush aggregate report.

Evaluation Modes

FLAIR separates task construction from scoring mode because continuous ICU prediction tasks can be evaluated in more than one way. Episodic scoring is used when there is one prediction per stay, as in extubation failure and ICU readmission. Landmark scoring is used for daily trajectories, as in T1 and T2; it reports performance at lead-time slices before the stay endpoint. Peak scoring is used for continuous screening tasks, as in T4; it collapses each stay to its highest predicted risk before computing discrimination and subgroup metrics. Calibration and decision-curve analysis are omitted for peak mode because taking the maximum of many probabilities distorts probability calibration. This distinction is load-bearing: a row-level metric can overstate the performance a unit would experience when acting on one alarm per stay.

Leaderboard Mechanics

For each task and site, FLAIR reports AUROC, AUPRC, and threshold-dependent metrics with bootstrap confidence intervals in `discrimination.json`. Calibration is summarized by slope, intercept, Brier score, and reliability points in `calibration.json` when the report mode supports calibration. Decision-curve analysis is written to `dca.json`. Subgroup performance is written to `fairness.json` across available sex, race, ethnicity, and age strata after suppression. Landmark tasks also write lead-time summaries. The leaderboard ranks submissions on the primary mode-matched metric for each task and displays per-site variation prominently. A single pooled AUROC can hide the failure mode FLAIR exists to surface.

An Open, Extensible Task Catalog

The five current tasks are the seed of a growing catalog, not a permanent ceiling. Any researcher or clinician can propose a new task as a single Python module in the FLAIR tasks directory containing a `META` dictionary and a `build()` function returning the canonical cohort frame. Acceptance review verifies reproducibility against MIMIC-CLIF, clinical face validity through clinician-led review, and site-governance acceptance at a minimum of two consortium sites before the task enters the federated suite. This is the structural counter to benchmark saturation: the catalog can grow with clinical need and community contribution rather than ossifying around one author group's choices (**??**). **TODO:** Confirm final task-review committee membership and voting process.

5 Federated Protocol

The federated protocol is what makes FLAIR's privacy properties operational. The actors are explicit. A **Model Owner** submits a frozen model, code folder, or container. The **Benchmark Committee**, composed of clinicians, methodologists, and other governance

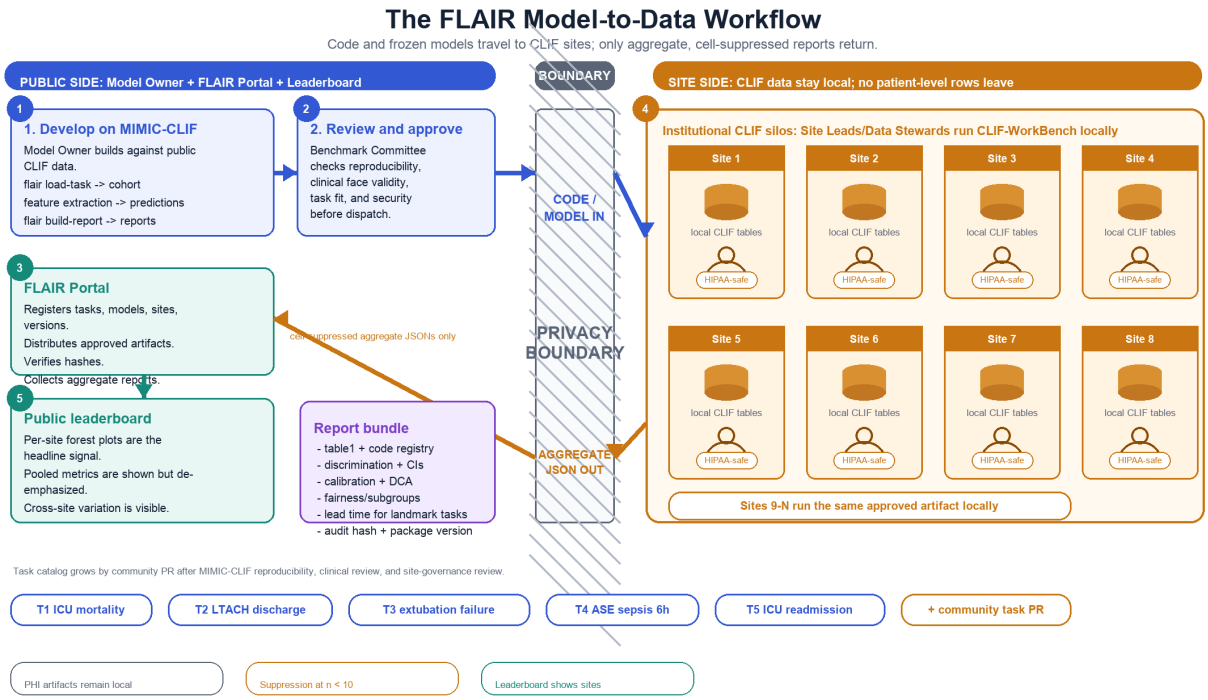


Figure 1: The FLAIR model-to-data loop. A Model Owner develops against MIMIC-CLIF and submits a frozen model, code, or container. The Benchmark Committee reviews the task or model for reproducibility, clinical face validity, and security. The FLAIR Portal registers the approved artifact and distributes byte-identical code to participating CLIF sites. Each Site Lead or Data Steward runs the artifact locally in CLIF-WorkBench against site CLIF tables. Patient-level cohorts, features, and predictions remain inside the institutional boundary. Only aggregate, cell-suppressed JSON report bundles return to FLAIR. The leaderboard emphasizes per-site variation, including forest-plot style displays, rather than only a pooled scalar. Source: authors.

participants, reviews task definitions and submissions for reproducibility, clinical face validity, and security concerns. **TODO:** Confirm whether regulators and patient representatives should be listed as standing committee roles. The **FLAIR Portal** registers tasks, models, sites, and versions; distributes approved artifacts; collects aggregate JSON reports; and renders the leaderboard. At each institution, a **Site Lead or Data Steward** runs the approved artifact against local CLIF tables and returns only the standardized non-PHI report bundle.

Submission

A model owner develops against MIMIC-CLIF. The library exposes CLIF tables through `clifpy`. The benchmark CLI exposes `flair load-task` to build task-specific cohorts and `flair build-report` to score predictions into report bundles. A submission may be raw code, a frozen model folder, or a container image that wraps both. Before any model is dispatched to a site, the Portal or review workflow checks that the submission runs against MIMIC-CLIF and follows the prediction-file contract. This model-to-data pattern is not new; it follows the lineage of DREAM Challenges and EHR DREAM (**??**). FLAIR's contribution is to turn that competition-style pattern into a standing ICU benchmark and submission service.

Execution

Each site runs the committee-reviewed, security-approved artifact inside the local CLIF-WorkBench environment or equivalent site-controlled environment. The submitted code may read local CLIF tables, write local cohorts, features, and prediction files, and invoke the FLAIR report builder. Local PHI artifacts remain in site-controlled storage. The intended upload is the aggregate report bundle: cohort summaries, discrimination, calibration where applicable, decision curves where applicable, subgroup summaries, lead-time summaries for landmark tasks, a suppressed code registry, and audit metadata. Every JSON carries the task name, task script hash, FLAIR package version, generation time, report mode, split, number of stays, prevalence, and the operating threshold used for threshold-dependent metrics.

Aggregation, Leaderboard, and Lifecycle

The central aggregator combines site-level JSON reports for a specific task, model, and version. The leaderboard should show per-site results and per-site variance prominently, because cross-site variation is the evidence FLAIR is designed to expose. Pooled metrics

can be shown, but they should not be the only headline. For tasks where sites opt in, the same protocol can support prospective silent-trial evaluation and longitudinal monitoring, consistent with emerging guidance on preregistration and monitoring of clinical AI (**???**). Prospective evaluation is a future protocol extension, not a claim made by the current baseline results.

Comparison With Adjacent Federated Infrastructures

The closest precedents for federated evaluation are complementary rather than redundant. OHDSI/LEGEND standardizes network studies on OMOP and often shares queries or coefficients (**?**). MedPerf provides hosted-model benchmarking for medical imaging (**?**). FeTS established federated training for brain MRI segmentation (**?**). OpenSAFELY moves analytic code to national primary-care data (**?**). FLAIR contributes the combination needed for ICU EHR-based clinical decision support: CLIF as a critical-care schema, model/code execution at sites, a hosted submission workflow, aggregate-only reporting, and a PR-extensible task catalog.

Feature	**OHDSI/ LEGEND**	**MedPerf**	**FeTS**	**Open SAFELY**	**FLAIR**
Data standard	OMOP	task specific	imaging	TPP/EMIS	CLIF
Federation unit	SQL/coefs	container	gradients	analytic code	model/code execution
Main use	network studies	imaging benchmarks	federated training	secure analysis	ICU EHR benchmarks
Hosted-model portal	no	yes	yes	no	yes
Patient rows shared	no	no	no	no	no
Extensible tasks	limited	partial	no	not applicable	yes

Table 4: FLAIR compared with adjacent federated or model-to-data infrastructures. The table is intended as orientation, not a formal feature ranking. Source: authors.

6 Toward the Evidence Expected for EHR-Based CDS

United States Food and Drug Administration guidance on clinical decision support software and the International Medical Device Regulators Forum framework for software as a medical device ask developers to make the basis of recommendations transparent and to demonstrate validation in the populations of intended use (**??**). Historically, that posture has been bottlenecked by the same access constraints described above: evidence is generated one bilateral data agreement at a time (**???**).

FLAIR's contribution to that posture is structural rather than declarative. Byte-identical task code and audit hashes make the cohort definition, exclusions, and outcome label re-inspectable across sites. Per-site discrimination, calibration, subgroup, and cohort JSONs create a mechanically auditable evidence package. Running across heterogeneous CLIF sites without patient-data movement provides a way to observe behavior under realistic distribution shift. FLAIR is not a regulatory submission, not a substitute for prospective clinical trials, and not a guarantee of clinical utility. It is an evidence layer between local model development and the regulator- or clinician-facing claims a developer may eventually want to make.

7 Discussion

FLAIR combines three properties that have appeared separately but not together at the scale ICU clinical AI requires: evaluation across heterogeneous sites without patient-data movement, a clinically scoped ICU data format that makes code portable, and a governance protocol that lets the task catalog grow through community contribution.

Saturation resistance through siloed data. Because FLAIR's private test data remain at participating sites, the data underlying the leaderboard do not enter public hands. This limits the risk that future models are pretrained directly on the evaluation set and reduces

the incentive for developers to tune to patient-level private test data they cannot see. FLAIR does not eliminate benchmark gaming or distribution shift, but it makes overfitting to a single public test set less central to the evaluation process.

Why evaluation, not training, federates first. FLAIR federates evaluation rather than training. That distinction matters. Aggregate metrics leak less patient information than model updates exchanged in federated learning, and the reciprocal risk of exposing a model to sites is bounded by committee review, compatibility checks, and local site approval. Only aggregate, cell-suppressed metrics leave a site. There is no per-site trained model object to share centrally. Evaluation-only federation is also a practical stepping stone toward future federated learning: it establishes a shared schema, code-to-data operations, templated governance, and site trust before any heavier training protocol is attempted.

Limitations. FLAIR depends on CLIF adoption. Sites that have not mapped their data to CLIF need an extract-transform-load step before they can participate. The current benchmark suite is limited to structured EHR data and binary classification. It does not include notes, waveforms, images, treatment-effect estimation, or prospective bedside impact. Aggregate-only reporting protects privacy but limits fine-grained error analysis. Suppressed small subgroups are not visible in central reports. The current protocol evaluates frozen models; it does not perform federated training. The v2 baseline numbers are useful for manuscript development but should be refreshed after task definitions are frozen, particularly for T4.

Roadmap. Near-term work should expand site participation, regenerate baselines after final task-code freeze, add site-level columns beyond Rush, and support opt-in prospective silent-trial evaluation. Additional CLIF tables, including diagnoses and procedures, can strengthen future tasks as the schema matures. The same model-to-data contract could later extend to notes, waveforms, and multimodal ICU data if those modalities are standardized

within or alongside CLIF. Federated learning is a feasible longer-term extension, but it would require stronger security guarantees, live site-to-server coordination, and separate institutional review.

Conclusion. FLAIR is a federated benchmark protocol for ICU prediction research. It combines a shared critical-care data format, public development on MIMIC-CLIF, five initial benchmark tasks, committee-reviewed model-to-data execution, and aggregate-only site reporting. Its main contribution is not a new model, but a reproducible way to ask whether ICU prediction models behave similarly across institutions.

Required Statements

Data and Code Availability. The FLAIR source code, task definitions, feature extractors, report-generation library, and v3 manuscript build scripts are released or intended for release under Apache 2.0 at `https://github.com/clif-consortium/FLAIR`. MIMIC-CLIF is distributed through PhysioNet under the applicable MIMIC data use agreement. Consortium site data are not redistributable and are accessible only through approved local execution. Baseline model archive DOI: **TODO:** DOI. Container registry tag: **TODO:** tag.

Institutional Review. Each participating consortium site obtained independent institutional review board approval for CLIF participation and local execution of benchmark code. Central protocol coordination is provided by the University of Chicago. **TODO:** Confirm final IRB wording for all sites.

Funding. TODO: grants.

Author Contributions. TODO: Fill CRediT taxonomy roles.

Conflicts of Interest. TODO: disclosures.

AI-Use Disclosure. TODO: Name LLM(s), if any, and identify the sections or artifacts they helped produce.

References Note. References are maintained in `docs/paper/references.bib`. The DOCX builder emits a numbered reference list for cited sources.

A Supplementary Appendix

S1. Count Feature Baseline

The baseline converts CLIF event rows into token counts. Events include vitals, labs, medications, respiratory support, patient assessments, and other configured domains. For each prediction row, only events before the prediction time are counted. The result is a sparse matrix with one row per prediction and one column per token. Missing tokens are zero. No task-specific feature curation is used in the baseline.

S2. Full MIMIC-CLIF Baseline Metrics

Task	Mode	n	Prev.	AUROC (95% CI)	AUPRC	Cal. slope	Cal. int.	Brier
T1, ICU daily mortality	landmark	18,195	0.134	0.864 (0.856–0.872)	0.587	1.19	-1.24	0.111
T2, LTACH discharge	landmark	18,195	0.065	0.789 (0.775–0.803)	0.288	1.01	-1.77	0.096
T3, extubation failure 24 h	episodic	6,693	0.054	0.696 (0.665–0.719)	0.134	0.45	-1.86	0.070
T4, sepsis 6 h	peak	19,943	0.063	0.693 (0.678–0.709)	0.134	—	—	—
T5, unplanned ICU readmission	episodic	16,806	0.093	0.641 (0.626–0.658)	0.248	0.68	-1.89	0.151

Table S1: Full v2 MIMIC-CLIF internal baseline metrics for the XGBoost count-feature model. Calibration is omitted for T4 peak scoring by design. Source: FLAIR v2 baseline artifacts. **TODO:** Regenerate after final task-code freeze.

S3. Supplementary Figures

The existing v2 report figures are retained as supporting material for v3. These include ROC overlays, calibration plots, decision-curve plots, landmark lead-time plots, and subgroup AUROC plots. **TODO:** Regenerate figures after final baseline rerun and update captions if T4 changes.